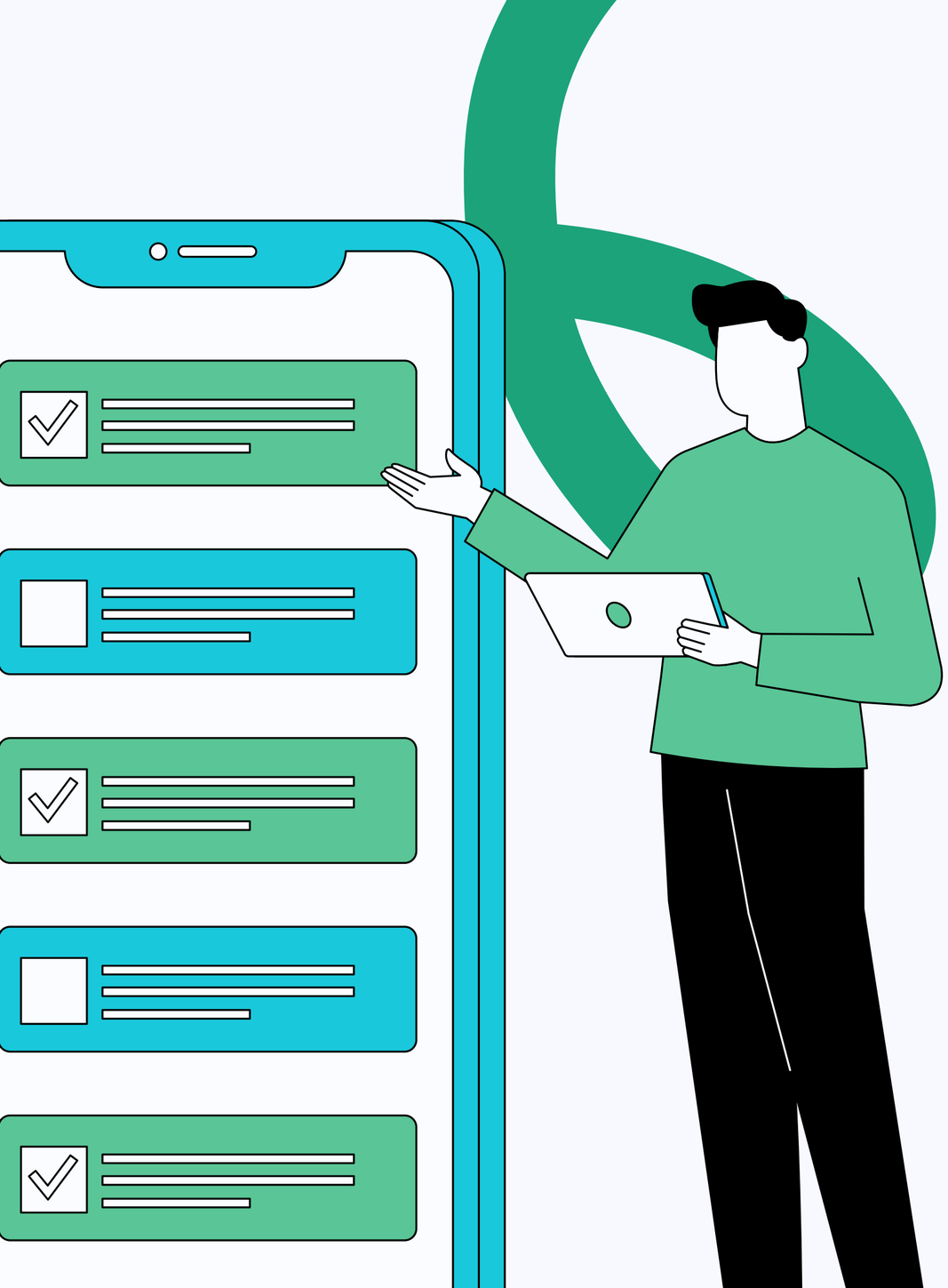
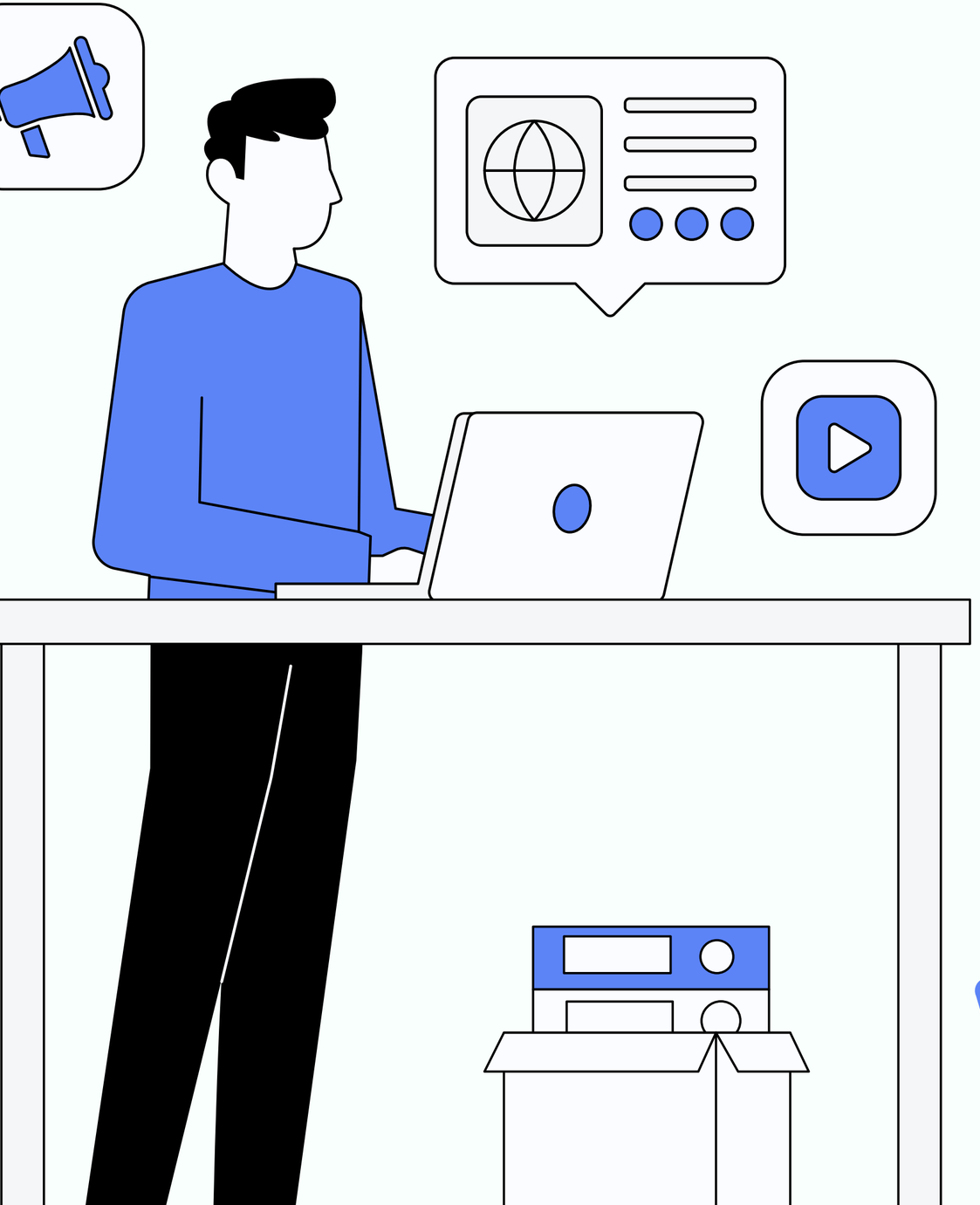


Evaluasi Tengah Semester [ETS]

Mata Kuliah Komputasi Awan

Implementasi End-to-End Data Pipeline pada Platform AWS untuk Deteksi Berita Hoaks dalam Skala Cloud Computing

[Kelompok 05]



Deteksi Berita Hoaks Berbasis Cloud (AWS)

Membangun sistem otomatis dari (end-to-end pipeline) infrastruktur AWS untuk mengidentifikasi apakah sebuah informasi atau berita termasuk hoaks atau fakta, dengan memanfaatkan dataset berita.

Alur dibagi menjadi tiga pilar utama terintegrasi di AWS:

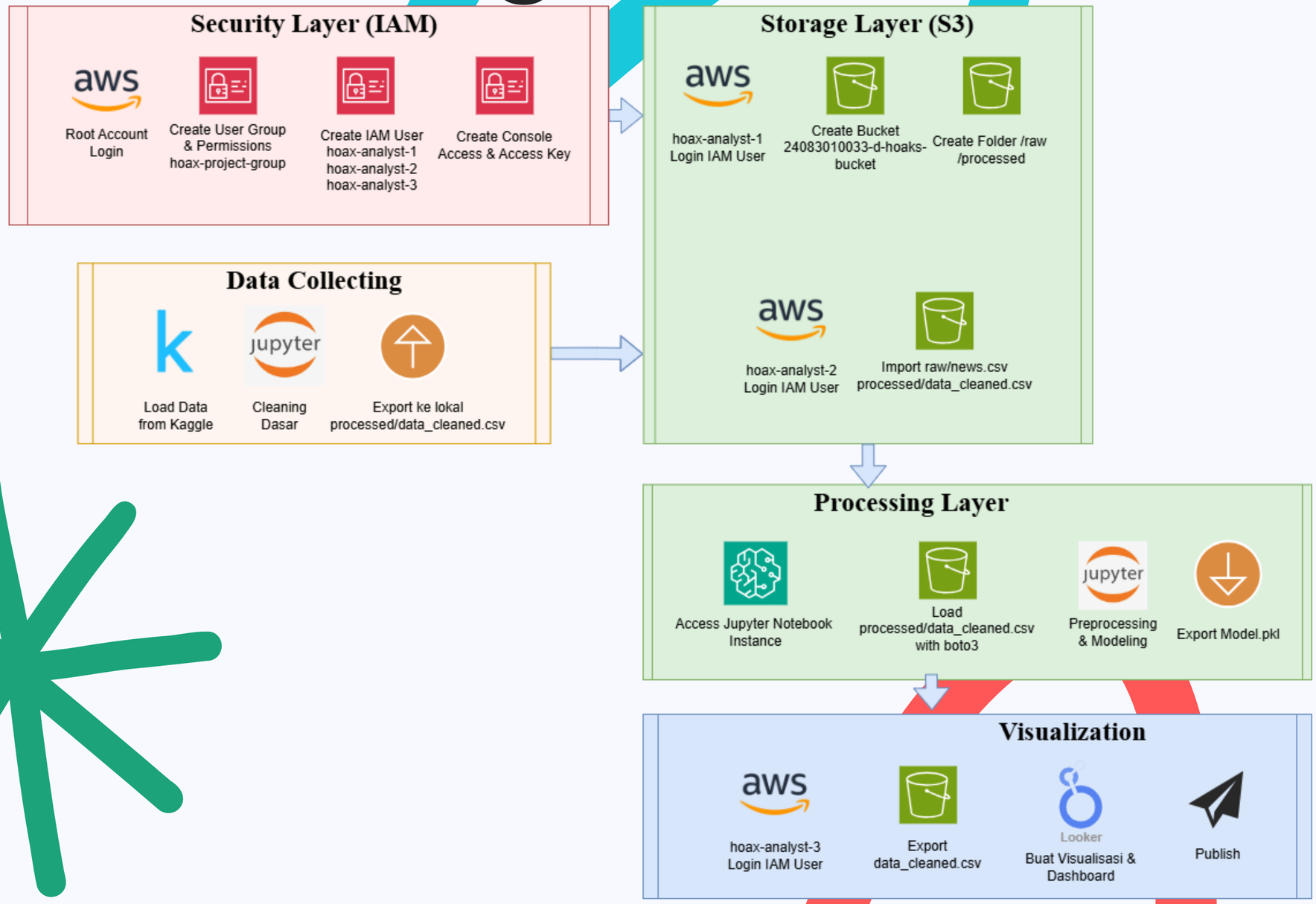
1. Penyediaan Infrastruktur (Foundation): "Rumah" data di AWS S3 dengan keamanan ketat melalui IAM User. **[Alifah 033]**

2. Pipeline Data & Cloud Processing: Dengan Amazon SageMaker sebagai mesin pemroses. Data dengan library Boto3, dibersihkan, dan diproses otomatis dalam cloud environment. **[Nada 066]**

3. Analisis, Modeling & Visualisasi (Output):

- Membuat Agregasi Performa Model
- Visualisasi: Mengubah data mentah menjadi wawasan visual **[Amalia 034]**

Arsitektur Diagram



Setup IAM

hoax-project-group Info

Summary

User group name	Creation time	ARN
hoax-project-group	April 13, 2026, 20:51 (UTC+07:00)	arn:aws:i -project-

Users (3) | Permissions | Access Advisor

Permissions policies (9) Info

Simulate | Remove

You can attach up to 10 managed policies.

Search | Filter by Type: All types

Policy name	Type
AmazonDataZoneFullAccess	AWS managed
AmazonEC2ReadOnlyAccess	AWS managed
AmazonS3FullAccess	AWS managed
AmazonSageMakerFullAccess	AWS managed
CloudWatchLogsFullAccess	AWS managed
IAMFullAccess	AWS managed
IAMSelfManageServiceSpecificCredentials	AWS managed
IAMUserChangePassword	AWS managed
SageMakerStudioAdminIAMConsolePolicy	AWS managed

Users (6) Info

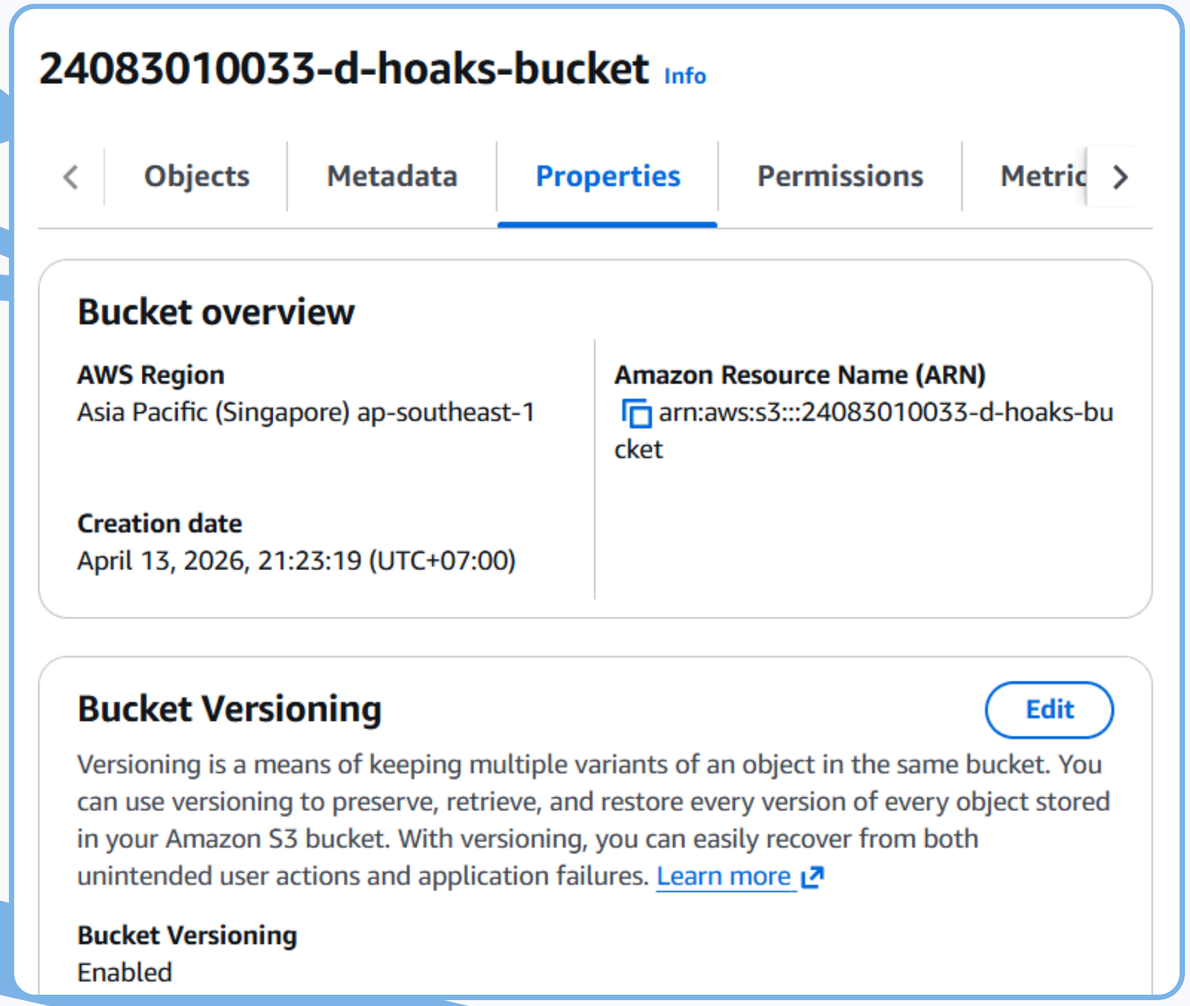
Delete | Create user

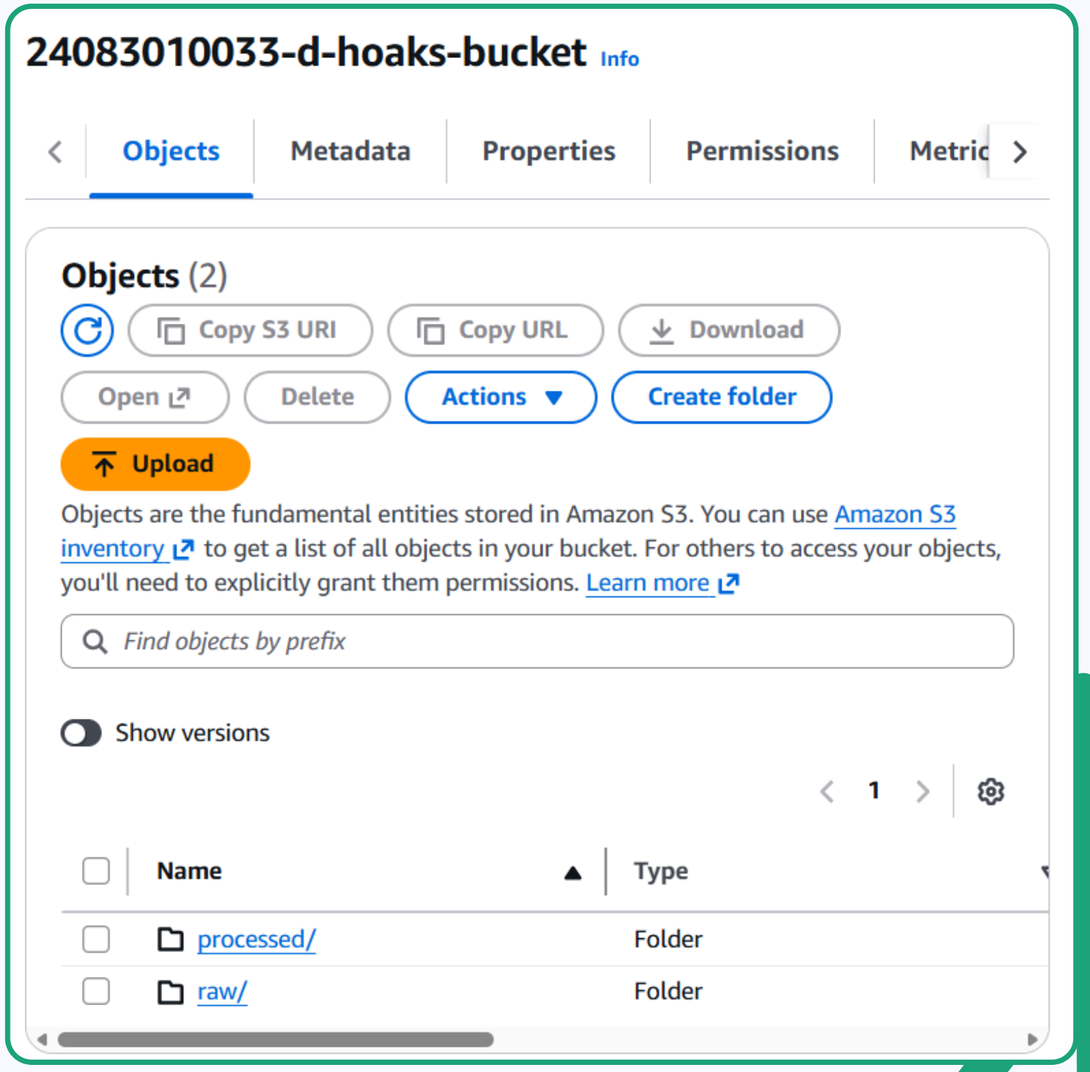
An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.

Search | 1

User name	Path	Groups	Last activity
hoax-analyst-1	/	1	10 hours ago
hoax-analyst-2	/	1	3 hours ago
hoax-analyst-3	/	1	4 hours ago

Setup Bucket





Cleaning Data

```python
# preprocessing dasar
X = data['text'].astype(str)
y = data['label'].map({'FAKE': 1, 'REAL': 0})
```

```python
cleaned_data = pd.DataFrame({
    "text": X,
    "label": y
    })

cleaned_data.head()
```

	text	label
0	Daniel Greenfield, a Shillman Journalism Fello...	1
1	Google Pinterest Digg Linkedin Reddit Stumbleu...	1
2	U.S. Secretary of State John F. Kerry said Mon...	0
3	— Kaydee King (@KaydeeKing) November 9, 2016 T...	1
4	It's primary day in New York and front-runners...	0

```python
cleaned_data.to_csv("../data/processed/data_cleaned.csv", index=False)
print("Saved cleaned data to data_cleaned.csv")
```

```
Saved cleaned data to data_cleaned.csv
```

```python
load_dotenv()
s3 = boto3.client(
    's3',
    aws_access_key_id=os.getenv("AWS_ACCESS_KEY_ID"),
    aws_secret_access_key=os.getenv("AWS_SECRET_ACCESS_KEY"),
    region_name=os.getenv("AWS_DEFAULT_REGION")
)

bucket_name = os.getenv("S3_BUCKET_NAME")
key = "processed/data_cleaned.csv"

csv_buffer = StringIO()
data.to_csv(csv_buffer, index=False)

s3.put_object(Bucket=bucket_name, Key=key, Body=csv_buffer.getvalue())
print("Upload ke S3 berhasil!")
```

```
Upload ke S3 berhasil!
```

Preprocessing

In [21]:

```
load_dotenv()
s3 = boto3.client(
    's3',
    aws_access_key_id=os.getenv("AWS_ACCESS_KEY_ID"),
    aws_secret_access_key=os.getenv("AWS_SECRET_ACCESS_KEY"),
    region_name=os.getenv("AWS_DEFAULT_REGION")
)

bucket_name = os.getenv("S3_BUCKET_NAME")
key = "processed/data_cleaned.csv"

s3.download_file(bucket_name, key, "../data/processed/data_cleaned.csv")
```

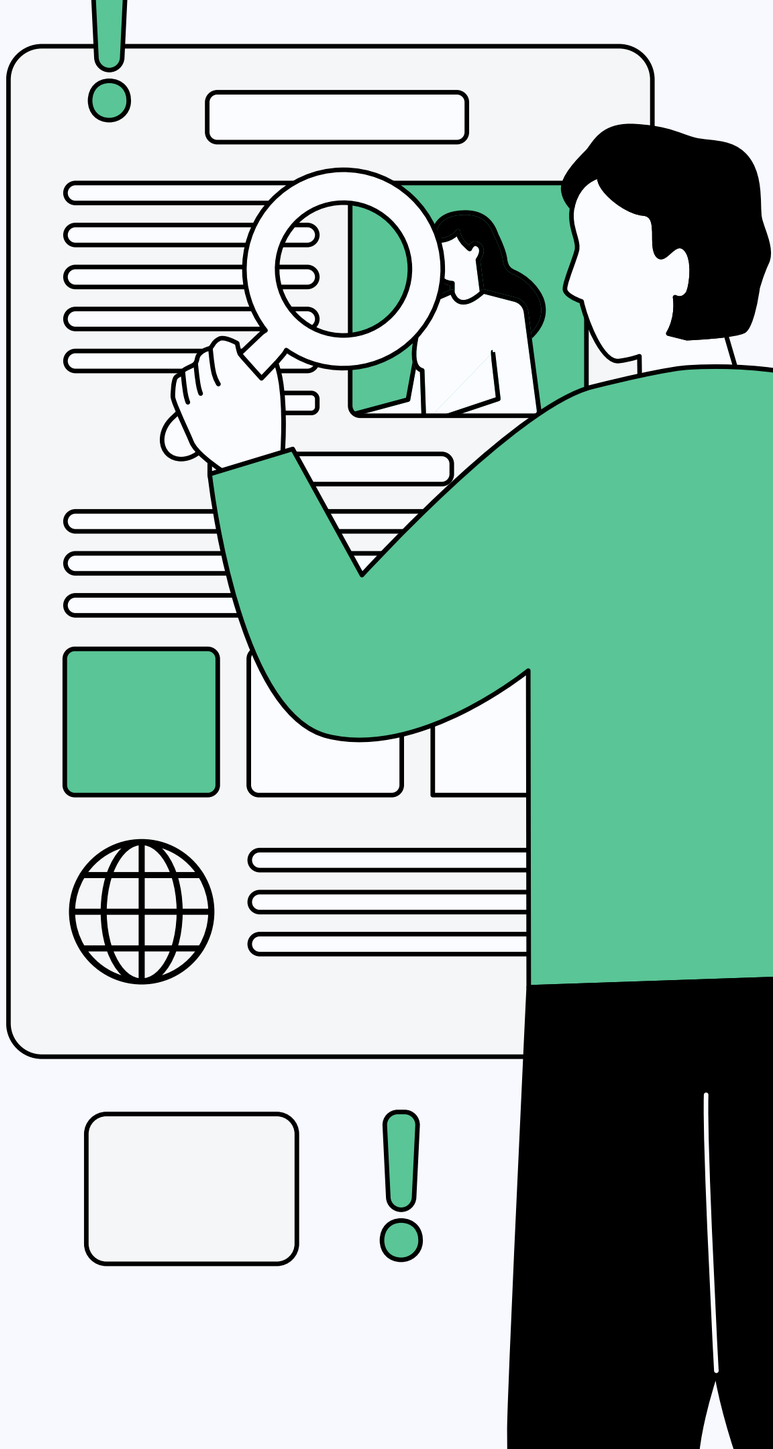
```
X = data['text']
y = data['label']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, stratify=y, random_state=42
)
```

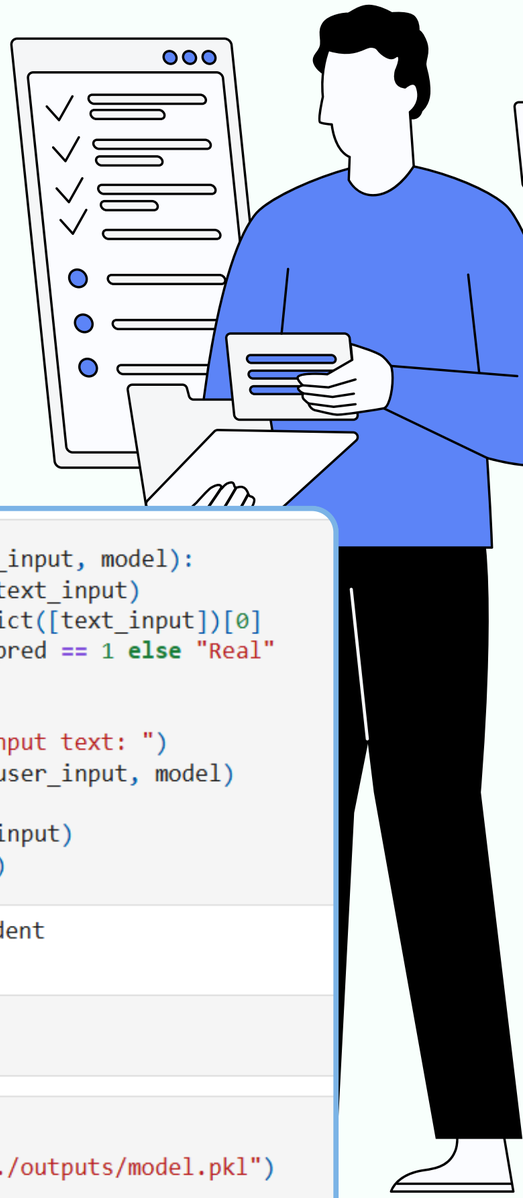
```
def tokenize(text):
    return text.lower().split()

sentences = [tokenize(text) for text in X_train]
print(sentences[0])
```

```
['president', 'obama', 'offered', 'enthusiastic', 'support', 'for', 'hillary', 'clinton', 'a
ainted', 'a', 'hopeful', 'picture', 'of', 'the', 'country.', 'u.s.', 'president', 'barack',
'clinton', 'wave', 'to', 'the', 'crowd', 'after', 'the', 'president', 'spoke', 'at', 'the',
'july', '27,', '2016.', 'president', 'obama', 'described', 'an', 'optimistic,', 'hopeful', '
he', 'democratic', 'national', 'convention,', 'pointedly', 'diverging', 'from', 'the', 'more
```



Modeling Data



```python
w2v_model = Word2Vec(sentences, vector_size=100, window=5, min_count=2)
```

```python
class Word2VecTransformer(BaseEstimator, TransformerMixin):
    def __init__(self, model):
        self.model = model
        self.dim = model.vector_size

    def fit(self, X, y=None):
        return self

    def transform(self, X):
        vectors = []
        for text in X:
            words = tokenize(text)
            vecs = [self.model.wv[w] for w in words if w in self.model.wv]
            vectors.append(np.mean(vecs, axis=0) if vecs else np.zeros(self.dim))
        return np.array(vectors)
```

```python
model = Pipeline([
    ("w2v", Word2VecTransformer(w2v_model)),
    ("clf", LinearSVC(class_weight='balanced'))
])

model.fit(X_train, y_train)
```

```
Pipeline(steps=[('w2v',
                 Word2VecTransformer(model=<gensim.models.word2vec.Word2Vec object at 0x0000023058AE52B0>)),
                ('clf', LinearSVC(class_weight='balanced'))])
```

```python
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

y_pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))
print("\nClassification Report:")
print(classification_report(y_test, y_pred, target_names=["Real", "Fake"]))
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))
```

```
Accuracy: 0.9029202841357538

Classification Report:
              precision    recall  f1-score   support

        Real       0.91      0.89      0.90       633
        Fake       0.89      0.91      0.90       634

    accuracy                           0.90      1267
   macro avg       0.90      0.90      0.90      1267
weighted avg       0.90      0.90      0.90      1267


Confusion Matrix:
[[565  68]
 [ 55 579]]
```

```python
def predict_text(text_input, model):
    text_input = str(text_input)
    pred = model.predict([text_input])[0]
    return "Fake" if pred == 1 else "Real"


user_input = input("Input text: ")
hasil = predict_text(user_input, model)

print("Input:", user_input)
print("Label:", hasil)
```

```
Input: Trump is a president
Label: Real
```

```python
# !pip install joblib
```

```python
import joblib
joblib.dump(model, "../outputs/model.pkl")
```

```
['../outputs/model.pkl']
```

Overview Dataset dan Statistik Utama

Jumlah Data Fake News	Jumlah Berita Asli	Jumlah Berita Hoax
6,335	3,171	3.2K

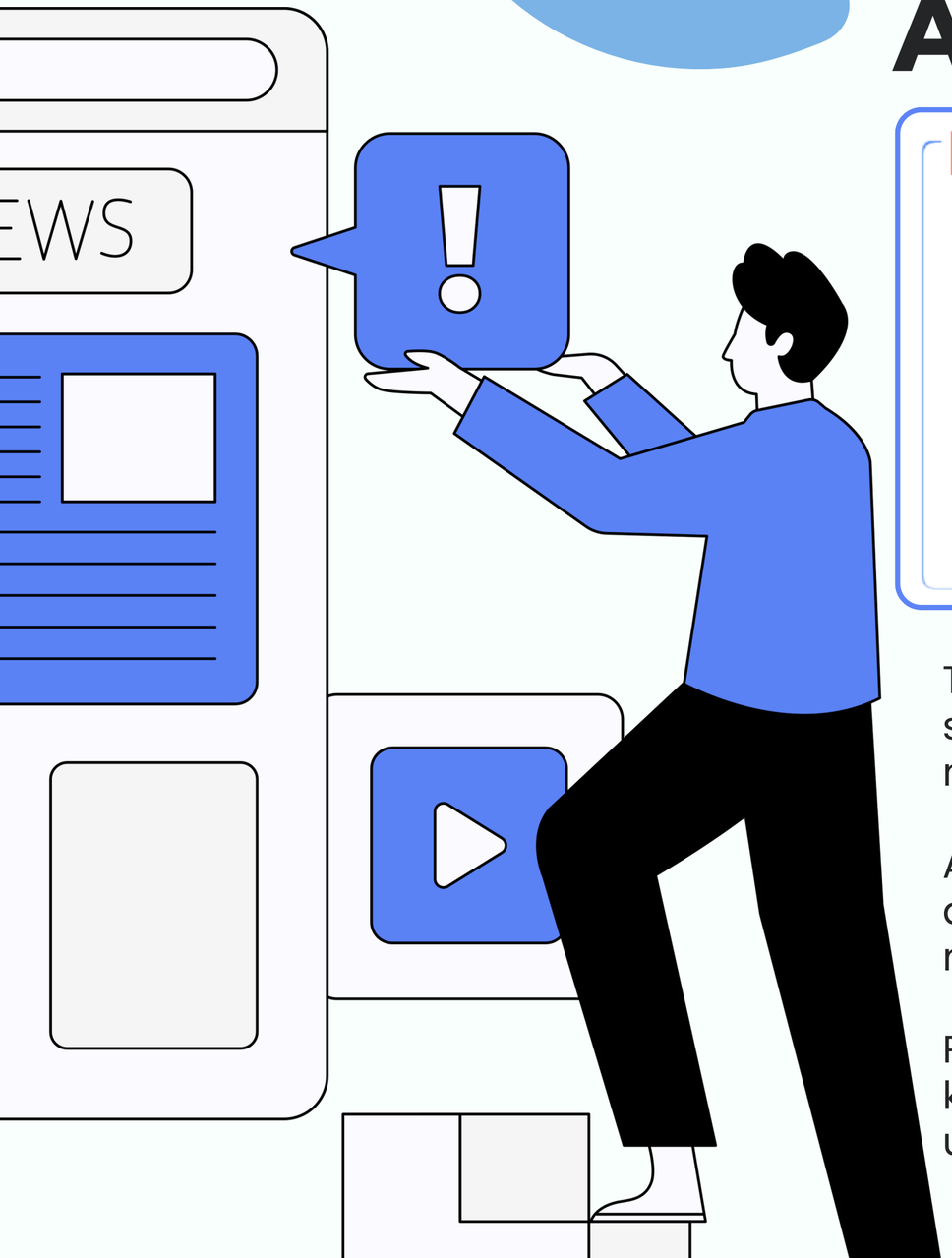
Skala Data: Total 6.335 entri data berita yang dianalisis.

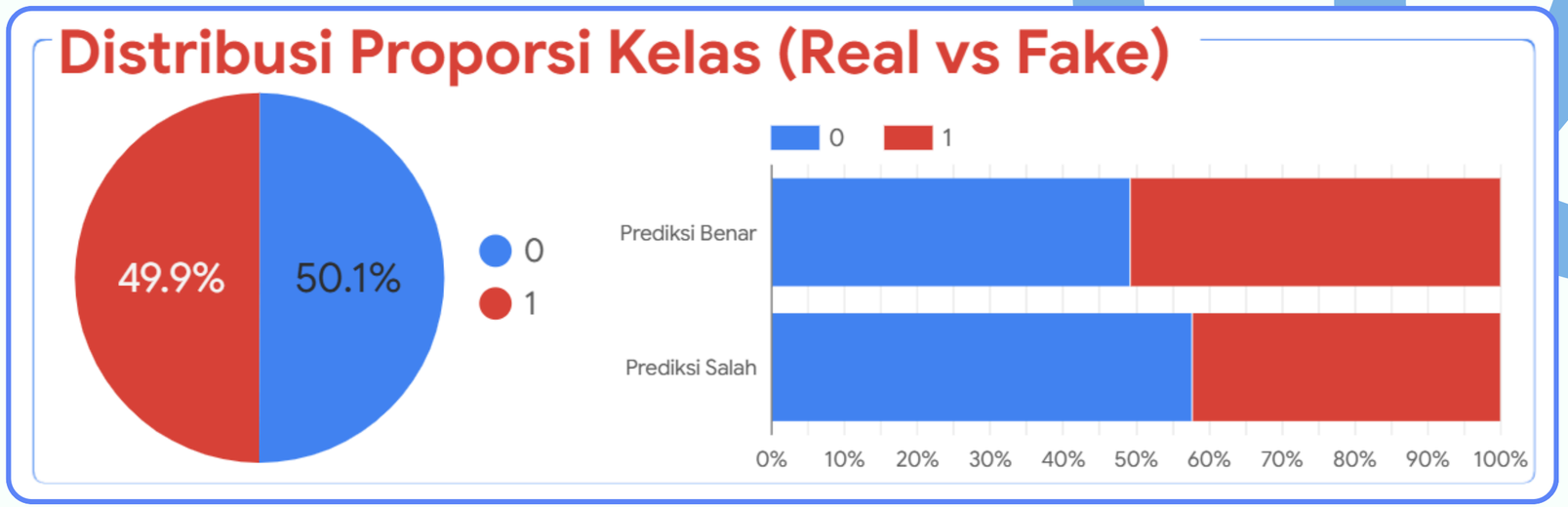
Komposisi Seimbang:

- 3.171 Berita Terverifikasi (Asli).
- ~3.200 Berita Terindikasi Manipulatif (Hoax).

High-Level Summary: Dataset mencakup berbagai topik berita dengan distribusi kelas yang hampir mencapai rasio 1:1.

Analisis Proporsi Kelas & Akurasi Prediksi





Terlihat bahwa Keseimbangan Data (Pie Chart) menunjukkan distribusi yang sangat sehat, yaitu 50,1% berita asli (Biru) dan 49,9% berita hoaks (Merah). Ini menjamin model tidak belajar secara memihak.

Akurasi Berdasarkan Kelas (Stacked Bar): Pada kategori Prediksi Benar, distribusi antara berita asli dan hoaks terlihat seimbang, menandakan model mampu mengenali kedua karakteristik dengan baik.

Pada kategori Prediksi Salah, terdapat kecenderungan model lebih sering keliru pada data berita asli (Biru lebih dominan). Hal ini menjadi catatan untuk pengembangan model di masa depan.

Eksplorasi Data Teks

Eksplorasi Data Teks

	Label	Isi Teks Berita	Panjang Teks
1.	1	Daniel Greenfield, a Shillman Journalism Fellow at the Freedom Center, is a New York writer focusing on radical Islam. In the final stretch of t...	7598
2.	1	Google Pinterest Digg Linkedin Reddit Stumbleupon Print Delicious Pocket Tumblr There are two fundamental truths in this world: Paul Ryan ...	2676
3.	0	U.S. Secretary of State John F. Kerry said Monday that he will stop in Paris later this week, amid criticism that no top American officials attend...	2567
4.	1	— Kaydee King (@KaydeeKing) November 9, 2016 The lesson from tonight's Dem losses: Time for Democrats to start listening to the voters. ...	2696
5.	0	It's primary day in New York and front-runners Hillary Clinton and Donald Trump are leading in the polls. Trump is now vowing to win enough ...	1840
6.	1	I'm not an immigrant, but my grandparents are. More than 50 years ago, they arrived in New York City from Iran. I grew up mainly in central N...	13469
7.	1	Share This Baylee Luciani (left), Screenshot of what Baylee caught on FaceTime (right) The closest Baylee Luciani could get to her boyfriend, ...	3221

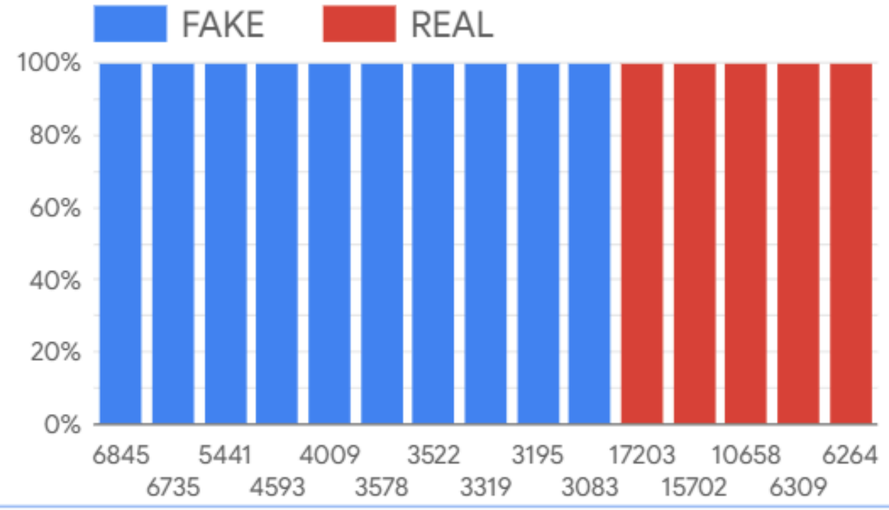
Input menampilkan sampel data mentah yang digunakan untuk melatih model, mencakup label (0/1), konten berita, dan metrik panjang teks.

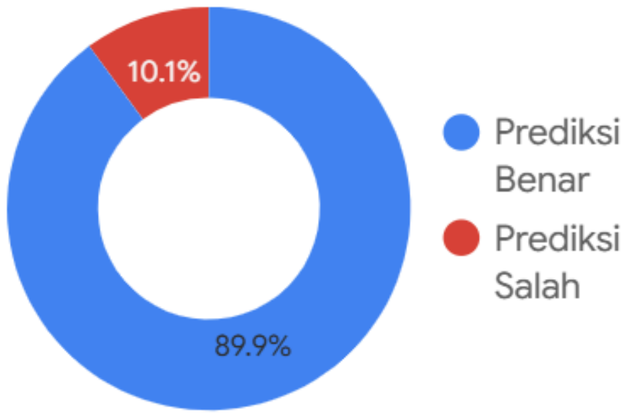
Karakteristik Fisik menunjukkan diversitas data, dari berita ringkas (1.800-an) hingga artikel mendalam > 13.000 karakter

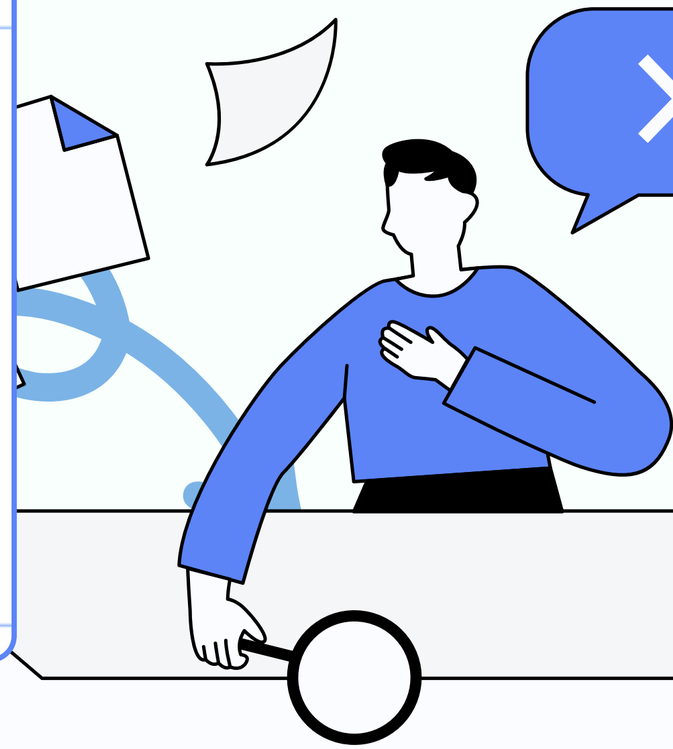
Data mencakup berbagai isu global dan politik, memberikan gambaran bahwa model dilatih dengan konteks dunia nyata yang kompleks, bukan data buatan.

Evaluasi Akhir: Performa Model & Karakteristik Data

Performa Model & Distribusi Karakter







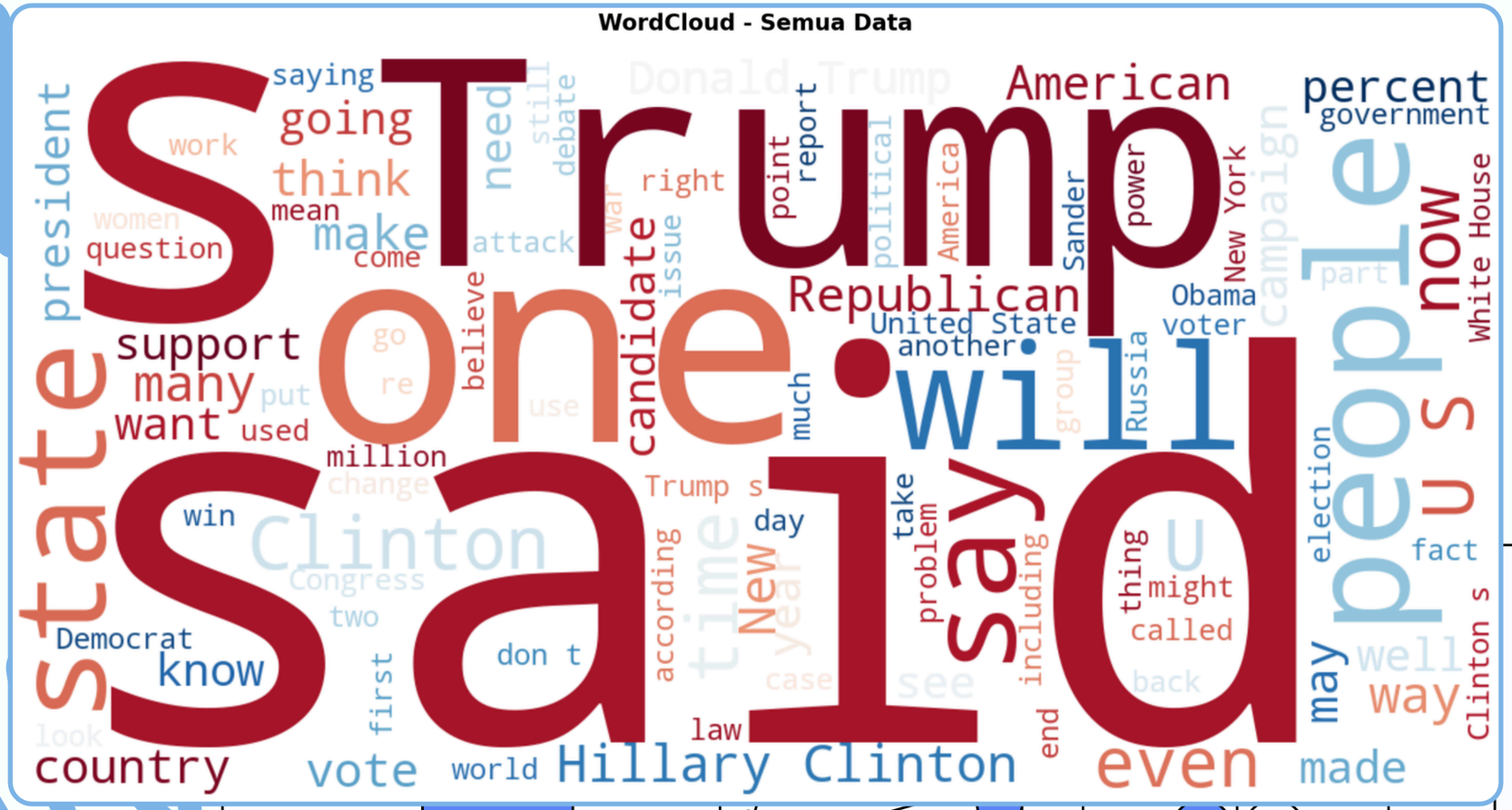
The Bottom Line (Akurasi), model berhasil mencapai tingkat akurasi sebesar 89,9%. Artinya, hampir 9 dari 10 berita berhasil ditebak dengan benar oleh sistem.

Distribusi Karakter: Melalui grafik batang, terlihat adanya pola unik di mana berita FAKE (Biru) cenderung mendominasi pada rentang panjang karakter tertentu dibandingkan berita REAL (Merah). Pola inilah yang diekstrak oleh model untuk membedakan keduanya.

Dashboard

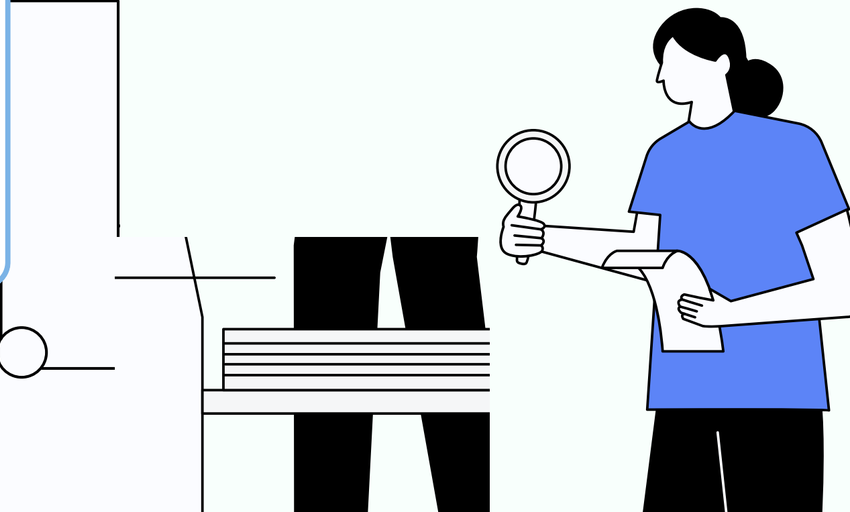
WordCloud untuk Semua Data

Terlihat bahwa data sangat didominasi topik politik Amerika Serikat, khususnya seputar Pemilu 2016, dengan munculnya 'Trump', 'Hillary', dan 'Clinton' yang berukuran sangat masif.



Topik Utama Dataset

- Fokus Tema: Politik AS (Pemilu 2016).
- Kata Tokoh Dominan: "Trump", "Hillary", "Clinton".
- Gaya Bahasa Jurnalistik: "said" (kutipan narasumber).
- Isu Kenegaraan & Spekulasi: Munculnya kata "state", "people", serta "will"



WordCloud

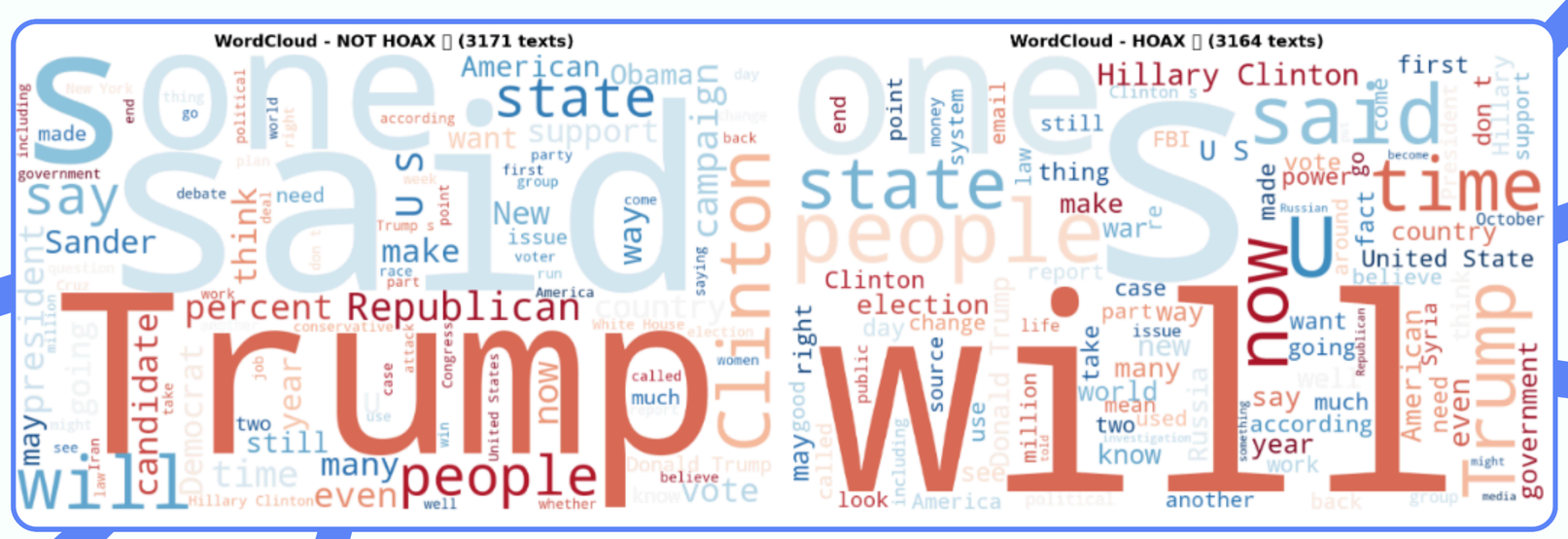
Perbandingan Karakteristik Teks: Asli vs. Hoaks

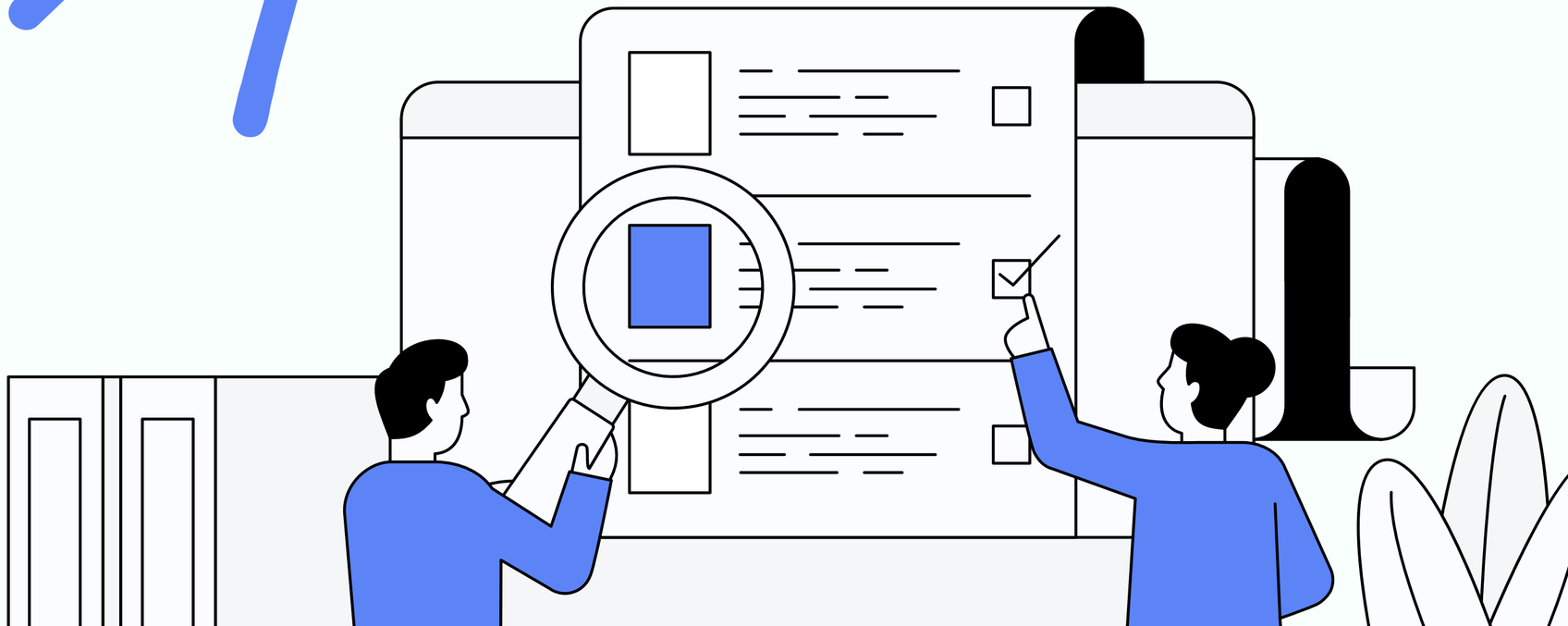
Kiri (NOT HOAX): Berita Asli

- Didominasi kata "said".
- Kutipan narasumber resmi dan faktual.
- Konteks politik lebih terstruktur ("State", "Republican").

Kanan (HOAX): Palsu

- Didominasi nama tokoh besar ("Trump", "Hillary").
- Indikasi taktik Clickbait (provokatif).
- Narasi spekulatif dan persuasif ("will", "people").





Thank You For Listening! ^^

Apakah ada pertanyaan? Feel free to ask us anything!